OPEN SUBSPACES OF THE REDUCTION OF A THICKENING LIFT UNIQUELY TO THE TOTAL SPACE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1648-1654 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 1656-1663, the complete author proof (6 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Uniqueness is stated in the lemma; the author's proof establishes existence via the site equivalence and openness of the lift.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The equivalence of big etale sites of a thickening (Theorem topological-invariance of the same chapter, collected as S3-spaces-topological-invariance L1411) and the criterion that an etale universally injective morphism is an open immersion (Morphisms of Spaces); cited inputs with the construction written in the retained span.

Source reference. Original lemma 05ZL: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1545–1547. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 05W5: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1663–1663. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $X \subset X'$ be a thickening of algebraic spaces over S. For any open subspace $U \subset X$ there exists a unique open subspace $U' \subset X'$ such that $U = X \times_{X'} U'$.*

Proof. Let $U' \to X'$ be the object of $X'_{spaces,étale}$ corresponding to the object $U \to X$ of $X_{spaces,étale}$ via (05ZL). The morphism $U' \to X'$ is étale and universally injective, hence an open immersion, see Morphisms of Spaces, Lemma 05W5. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.